

# **MENG 223 Mechanical Design I Course Project Report**

## **Design Constraints and Requirements Report**

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### **Team Name:**

MechalInnovators

### **Team Members:**

- |                             |           |
|-----------------------------|-----------|
| 1. Amro Maged Mohammed      | 231001357 |
| 2. Moamen Ahmed Rushdy      | 231001011 |
| 3. Ahmed Saleh Mohamed      | 231001847 |
| 4. Saif Abdelmoamen Mohamed | 231001990 |
| 5. Osama Walid              | 221002290 |
| 6. Ahmed Mohamed Afifi      | 231001757 |

Under supervision of

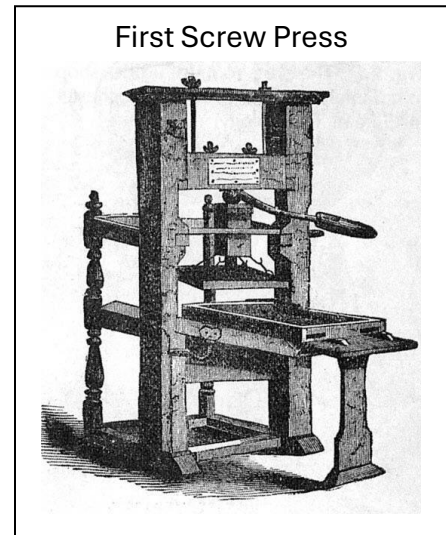
**Prof. Ahmed Sakr**

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## **Study of Existing Presses and Their Components**

The first press machine developments were in ancient times when human power and primitive tools were used to shape the materials. One of the earliest presses to be known to man was invented by the Greeks in the 1st century AD and employed the use of the screw press. It was commonly used to treat the pressing of grape juice and olive oil for winemaking. The screw press was later improved in design by the Romans and continued to be used for a vast array of applications, including printing books in the 15th century through Johannes Gutenberg's printing press.



Mechanical press machines powered by flywheel and crankshaft were invented with the Industrial Revolution (18th-19th century) to enhance force and efficiency. Hydraulic presses were invented, initially by Joseph Bramah in 1795. Hydraulic presses became a norm for manufacturing industries, especially metal forming, forging, and deep drawing.

In the 20th century, electric and pneumatic presses had been available, with faster and more accurate control. Modern servo-electric presses today drive computer-aided manufacturing with economy, accuracy, and programmability. Presses developed to cope with growing industry demands, ranging from food processing to automotive and aeronautical engineering, year by year.

Screw presses are mechanical devices widely utilized in industries such as manufacturing, metalworking, food processing, and waste management. They are primarily employed for tasks like metal forming, stamping, compression, extrusion, and briquetting. This study delves into the design, functionality, and components of screw presses, providing a comprehensive understanding of their operation and applications.

## 1. Power Screw

The **power screw** is the core component of a screw press, responsible for converting rotational motion into linear motion. It applies the necessary force to the workpiece to achieve the desired deformation or compression.

- **Types of Power Screws:**
  - **Lead Screws:** Commonly used for precision applications due to their high accuracy.
  - **Ball Screws:** Offer low friction and high efficiency, suitable for high-speed operations.
  - **Acme Screws:** Known for their durability and ability to handle heavy loads.
- **Material:** Typically made from high-strength steel or alloy to withstand high stress and wear.
- **Thread Design:** The thread profile (e.g., square, trapezoidal, or triangular) determines the efficiency and load-bearing capacity of the screw.
- **Applications:** Used in metal forming, forging, and extrusion processes where controlled force is required.

## 2. Bolts

**Bolts** are critical for securing the structural components of the press, ensuring stability and integrity under operational loads.

- **Types of Bolts:**
  - **Hex Bolts:** Commonly used for general-purpose fastening.

One of the largest presses



- **High-Strength Bolts:** Used in heavy-duty applications to withstand high stress.
- **Anchor Bolts:** Secure the press to the foundation or floor.
- **Material:** Typically made from carbon steel, alloy steel, or stainless steel, depending on the load and environmental conditions.
- **Torque Requirements:** Proper torque application is essential to prevent loosening during operation.
- **Safety Considerations:** Bolts must be regularly inspected for wear, corrosion, or fatigue to avoid structural failure.

### 3. Springs

**Springs** in screw presses serve as energy-absorbing components, providing the necessary force during the pressing operation and ensuring smooth operation.

- **Types of Springs:**
  - **Compression Springs:** Absorb energy by compressing under load.
  - **Torsion Springs:** Store energy through rotational force.
  - **Die Springs:** Specifically designed for high-load applications in presses.
- **Material:** Typically made from high-carbon steel or alloy steel for durability and resilience.
- **Function:** Springs help maintain consistent pressure, reduce vibration, and protect the press from sudden load changes.
- **Applications:** Used in stamping, punching, and forming operations to ensure uniform force distribution.

### 4. Belt Drive System

The **belt drive system** transmits power from the motor to the screw, enabling the rotational motion required for the press operation.

- **Types of Belts:**
  - **V-Belts:** Commonly used for their efficiency and ability to handle high torque.
  - **Timing Belts:** Provide precise synchronization between the motor and screw.
  - **Flat Belts:** Used in low-power applications.
- **Material:** Belts are typically made from rubber, polyurethane, or reinforced materials for durability.
- **Maintenance:** Regular inspection for wear, tension, and alignment is necessary to ensure optimal performance.

- **Advantages:** Belt drives are cost-effective, reduce vibration, and provide smooth power transmission.

## 5. Chassis and Frame

The **chassis and frame** form the structural backbone of the screw press, providing stability and support during operation.

- **Material:** Constructed from high-strength steel or cast iron to withstand heavy loads and resist deformation.
- **Design Considerations:**
  - **Rigidity:** Ensures minimal deflection underload.
  - **Modularity:** Allows for easy assembly, disassembly, and maintenance.
  - **Ergonomics:** Designed to facilitate operator safety and ease of use.
- **Applications:** The frame must be tailored to the specific requirements of the press, such as size, capacity, and operational environment.

## 6. Additional Features of Modern Screw Presses

Modern screw presses incorporate advanced features to enhance performance, safety, and usability:

- **Adjustable Speed Ranges:** Allow operators to control the speed of the press for different materials and applications.
- **Automation:** Integration with programmable logic controllers (PLCs) for precise control and repeatability.
- **Safety Mechanisms:** Include emergency stop buttons, safety guards, and overload protection to prevent accidents.
- **Lubrication Systems:** Automated lubrication systems reduce wear and extend the lifespan of components.
- **Energy Efficiency:** Modern designs focus on reducing energy consumption while maintaining high performance.

## 7. Applications of Screw Presses

Screw presses are versatile machines used in a wide range of industries:

- **Metalworking:** Forging, stamping, and extrusion of metals.
- **Food Processing:** Oil extraction, juice pressing, and dewatering.
- **Waste Management:** Compaction of waste materials for recycling or disposal.

- **Pharmaceuticals:** Tablet compression and powder compaction.
- **Ceramics and Composites:** Forming and shaping of materials under high pressure

8. Comparative Analysis of Existing Presses

A comparison of existing screw presses reveals variations in design, capacity, and functionality:

| Parameter        | Small-Scale Presses      | Medium-Scale Presses  | Large-Scale Presses |
|------------------|--------------------------|-----------------------|---------------------|
| Capacity         | Low (up to 50 tons)      | Medium (50–200 tons)  | High (200+ tons)    |
| Speed            | High                     | Moderate              | Low                 |
| Applications     | Light-duty tasks         | General-purpose tasks | Heavy-duty tasks    |
| Automation Level | Manual or semi-automatic | Semi-automatic        | Fully automatic     |
| Cost             | Low                      | Moderate              | High                |

9. Challenges and Future Trends

- **Challenges:**
    - Wear and tear of components due to high stress.
    - Maintenance requirements for optimal performance.
    - High initial investment for advanced models.
  - **Future Trends:**
    - Integration of IoT for real-time monitoring and predictive maintenance.
    - Development of lightweight, high-strength materials for components.
    - Increased focus on energy-efficient and eco-friendly designs
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# **Explanation of the Key Features and Specifications of a Press**

Screw press is a multi-purpose machine used in different industries such as metal working, forging, and sewage treatment. It must be strong, precise, and secure. Below is a detailed explanation of its primary features and specifications:

## **1. Motor Power**

The motor forms the heart of the screw press. It provides the power needed to drive the machine. Motor power determines the amount of pressure the press can deliver and at what speed it can work.

Why is it important: The bigger motor can take in heavier loads and harder work. For example, a light press for a workshop can have a 3-kW motor, whereas a massive industrial press for factories can support motors that are over 120 kW.

Example: When you work with heavy metal plates, you require a press that is fitted with a heavy-duty motor to provide enough pressure to mold the metal.

## **2. Speed Range**

Speed range is how fast the screw press can move. Most screw presses allow you to adjust the speed, and you should be able to do this because various jobs and materials require various speeds.

Why it matters: For accuracy-conscious work, you might want to work slowly to ensure precision. For faster work, you can go faster to save time. Example: If you are stamping a pattern onto thin metal, you might prefer to use a slow speed, so you don't damage the material. But if you are stamping numerous parts, you might prefer to increase speed to work more efficiently.

## **3. Bed Length**

Bed length is the width of the work area where you place the material or workpiece. It determines the type of pieces you can work on, either large or small.

Why is it important: The bigger bed allows you to work with longer or greater materials, and a small bed takes up less space and suits very tiny jobs.

Example: A small press with a bed length of 300 mm can be used for small metal components, while a large press with a bed length of 1800 mm or more can handle larger work sizes, e.g., metal plates.

#### **4. Tool Rest**

Tool rest is a firm based upon which you place tools or dies. It keeps everything firmly in position and makes it work.

Why it matters: A stiff tool rest prevents shaking and makes the workpiece get formed as it is supposed to. In the absence of a proper tool rest, the material would move around, resulting in errors or poor output.

Example: During forging a metal part, the tool rest maintains the die steady so that the metal gets formed just as needed.

#### **5. Ergonomics Design**

Ergonomics is all about making the machine comfortable and easy to use for the operator. Modern screw presses are designed with fatigue-reducing characteristics and simplifying the work to be less exhausting.

Why it matters: If the machine is comfortable and easy to use, the operator can work longer and more effectively without fatigue.

Features:

Easy-to-reach controls so that the operator does not have to stretch or bend uncomfortably.

Adjustable controls to suit the operator's preference.

Simple interfaces, e.g., buttons or touch screens, to make operating the machine easy.

#### **6. Safety Features**

Safety is probably the most important feature of any industrial machine. Screw presses incorporate several safety features to protect the operator and prevent accidents.

Why it matters: Operating heavy machinery poses risks, so safety features are necessary to avoid injury and harm to the machine.

### **Key Safety Features:**

**Emergency Stop Buttons:** These enable the operator to easily shut off the machine in case something goes amiss.

**Guards:** These enclose moving parts so that the operator will not unintentionally touch them.

**Overload Protection:** It stops the machine from running when excessive pressure is used, which can hurt the press or the workpiece.

### **Other Specifications**

In addition to the major specifications, screw presses also have other technical specifications that determine how they work:

#### **Force Capacity (Tonnage):**

The maximum force of the press, in tons.

Why it matters: To create heavy-duty applications, you require a press that has a high force capacity to mold strong or thick material.

#### **Ram Stroke:**

This is the distance that the ram (the moving piece) can move.

Why it matters: Having a longer stroke means you can handle thicker or taller materials.

#### **Screw Pitch:**

This refers to the interval between the threads on the screw. It impacts the precision of the press as well as how much pressure is possible.

Why it matters: The lower pitch (closer threads) has more precision, but the larger pitch (widely spaced apart) has faster movement.

#### **Automation:**

Most modern screw presses incorporate automatic features, like programmable controls or sensors, to enhance the efficiency of the process.

Why it matters: Automation speeds up the machine and reduces the work required, hence the process is efficient and less error prone.

## **Applications of Screw Presses**

Screw presses are applied in different industries for different reasons, such as:

**Forging:** Shaping metal pieces, e.g., tools or gears.

**Stamping and Forming:** Making patterns or shapes on materials such as metal sheets.

**Dewatering:** Driving out water from materials such as from food processing or wastewater.

These machines are very resilient, precise, and flexible to operate a wide range of different types of jobs, so they are very applicable in industry.

## **Design Requirements of a Small-Scale Screw Press**

A screw press is a type of mechanical press that applies a helical screw mechanism to generate high-pressure force to work various materials. A small-scale screw press design must adhere to certain constraints and requirements to be efficient, cost-effective, durable, and easy to use.

### **Critical Design Constraints and Requirements**

#### **1. Budget Constraints**

The total material and manufacturing costs must be within budget constraints. Some commercially employed machines are as follows:

- Mini Expeller Screw Press Oil Maker Machine: \$56,000
- Mild Steel Commercial Expeller Screw Press Machine: \$1,600,000
- Automatic Dewatering Screw Press Machine: \$280,000

The structure must be designed to minimize costs in material selection and manufacturing operations without compromising on quality.

#### **2. Space Requirements**

The press must be fit into a previously allocated space, so small-sized designs are best. Typical dimensions to refer to are:

- Mini Expeller Screw Press Oil Maker Machine: 350x160x360mm
- YZS-68 Mini Oil Expeller: 900x390x750mm

Modular parts for easy installation and relocation in small spaces should be considered in the design.

#### **3. Power Supply Considerations**

The press must be designed to work with standard power supply requirements:

- Mini Expeller Screw Press Oil Maker Machine: 220V
- Mild Steel Commercial Expeller Screw Press Machine: 415V

Energy-efficient motors must be employed for reduced operational expenses and increased sustainability.

#### **4. Load Capacity**

The press should be able to exert a force strong enough to perform the desired processing. Key capacity specifications are:

- Mini Expeller Screw Press Oil Maker Machine: 1200W
- Mild Steel Commercial Expeller Screw Press Machine: 56 kW
- YZS-68 Mini Oil Expeller: 5.5 kW

Screw mechanism and motor should be strong enough to resist the required load without excessive wear or breakdown.

#### **5. Safety Features**

For user safety, the design should be incorporated:

- Emergency stop features for instant shutdown.
- Overload protection against burnout and structural failure.
- Good shielding around moving parts to prevent accidental injury.

#### **6. Durability and Material Selection**

Structural components must be made of rugged materials:

- Cast Iron Bed: In large-scale applications as it provides vibration resistance.
- Stainless Steel Body: Provides corrosion resistance and durability.
- Mild Steel or Stainless-Steel Frame: Supports the structure while maintaining the weight minimum.
- High-Tensile Alloy Screws: Reduces wear and increases operating life.

#### **7. Ease of Maintenance**

The design should be repairable and maintainable with ease by the incorporation of:

- Modular assembly to easily disassemble and reassemble.
- Lubrication points to reduce friction and wear by making them accessible.

-Replaceable elements such as bearings and seals to ensure maximum machine lifespan.

## **Additional Design Considerations**

### **1. Efficiency and Performance Optimization**

- Gear Reduction System: Maximizes torque output while reducing power consumption.
- Variable Speed Control: Allows users to control processing speed based on material requirements.
- Heat Dissipation System: Prevents overheating during extended operation.

### **2. Noise and Vibration Control**

- Anti-vibration Mounts: Reduce operational noise and improve stability.
- Sound Insulation Panels: Minimize environmental noise pollution at the workplace.
- Precision Engineering: Avoids misalignment and useless friction, maximizes efficiency.

### **3. Environmental Conditions**

- Waste Management System: Grapes and processes left material with utmost efficiency.
- Energy-Efficient Motors: Save power consumption and carbon emission.
- Biodegradable Lubricants: Have minimal environmental effect compared to conventional oils.

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## **Definition of Design Constraints Based on Customer Requirements**

A local garage that performs auto body repair and part manufacturing. They need a **screw press** for small to medium-sized manufacturing workshops. Their goal is to create a **cost-effective, compact, and durable machine** that aligns with workshop needs while maintaining safety and efficiency.

### **Customer Profile:**

- **Industry:** Small and medium-sized manufacturing workshops
- **Budget:** Limited, requiring cost-efficient design solutions
- **Space Availability:** Small workshops with limited floor space
- **Power Supply:** Standard workshop electrical power
- **User Experience:** Operators with basic technical knowledge

### **Decision-Making Process:**

#### **1. Budget Consideration:**

- The use of **mild steel** for the frame instead of expensive alloys.
- They limit the use of electronic controls, opting for **manual or semi-automatic operation**.

#### **2. Space Management:**

- The press is designed with a **vertical layout** to save floor space.
- It features a **foldable worktable** to adapt to different workshop sizes.

### 3. Power Supply Constraints:

- The motor selection is optimized for **low power consumption** while maintaining required torque.
- Alternative hand-operated mechanisms are explored for areas with unstable electricity supply.

### 4. Safety Measures:

- Integrate a **transparent safety shield** to prevent accidental hand injuries.
- Design an **emergency stop button** near the operator's reach.
- The belt drive is enclosed in a **protective casing** to prevent entanglement.

### 5. Ease of Use Considerations:

- The **press handle is ergonomically designed** for minimal strain.
- A **single button start/stop mechanism** is included for easy operation.

### 6. Durability & Long-Term Reliability:

- The screw is made of **hardened steel with anti-corrosion coating**.
- The frame is reinforced to **withstand repeated heavy loads**.

### Benefits of the design:

- They can use the press for **straightening metal parts, pressing bearings, and assembling components**.
- It fits into a **crowded workshop** where space is at a premium.
- Mechanics can operate it **without needing extensive training**.
- The robust safety features prevent accidents in a busy work environment.